

12/09 Questions

1) Write a Program to print tables of given number

Sample o/p:

$$x * y = y$$

$$x * 2 = y$$

$x * 3 = y$
 $x * 4 = y$
 $x * 5 = y$

$$x * 10 = y$$

here given value multiplied by
 i value & the result means
 updated result got multiplied by updated
 i will get i value from here
 for each iteration.

Code:-
`num = int(input("Enter a no:"))`

`for i in range(1, 11, 1):`

`Print(f"{num} * {i} = {num * i}")`

o/p:- enter a no: 5

$$5 * 1 = 5$$

$$5 * 2 = 10$$

...

$$5 * 10 = 50$$

given no: 5
 i value = 5
 updated result stored in num
 $5 * 1 = 5$
 $5 * 2 = 10$

2) Write a Program to Print factorial of a given number

Sample o/p:-

5

$$5 * 4 * 3 * 2 * 1 = 120$$

Code:-
`num = int(input("Enter the number:"))`

`fact = 1`

`for i in range(num, 0, -1):`

`fact = fact * i`

`Print(fact)`

$$1 * 5 = 5$$

$$5 * 4 = 20$$

• 11:

enter the number: 5

5

20

60

120

120

Note: if you type Print Statement outside the for loop
it only prints 120

Explanation:-

Fact = 1 (this means we are starting our factorial calculation with the value 1)

- sets up an accumulator variable with 1
- we set fact = 1 because in multiplication, starting with 1 is safe.
- (if we started with 0, everything becomes 0)
- factorial is a product of no

Ex:- for 5! (factorial of 5):

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

to calculate this in a program we must keep multiplying numbers one by one.

If we don't start with 1, the multiplication won't work. Correct

If we start with fact = 1:

$$\text{fact} = 1 \times 5 = 5$$

$$\text{fact} = 5 \times 4 = 20$$

Correct answer

if we start with fact = 0:

$$\text{fact} = 0 \times 5 = 0$$

Then everything stays 0 → Wrong result

Note:

For addition, we start from 0

Ex: Sum = 0 then keep adding

For multiplication, we start from 1

Ex: fact = 1, then keep multiplying

fact = 1

fact = fact * i

fact = 1 * 5 = 5

fact = 5 * 4 = 20

fact = 20 * 3 = 60

i = 5, 4, 3, 2, 1

again this value stores in fact variable & keep updating until stop condition

• Before loop starts

fact = 1 (nothing multiplied yet)

• iteration 1:

→ i = 5

→ fact = 1 * 5 = 5

→ Print → 5

iteration 2:

→ i = 4

→ fact = 5 * 4 = 20

→ Print → 20

8) Write a Program to calculate sum of 1st 5 numbers from 1 to 5

Sum = 0

for i in range(1, 6, 1):

Sum = Sum + i

Print(Sum)

o/p: 15

~~Sum~~

Note:

Sum = Sum + i

Print(Sum)

o/p: 1
3
6
10
15

for

Sum = Sum + i

Print(Sum)

o/p: 15

4) Write a Program to calculate sum from 1 to given number

```
num = int(input("enter a no: "))
```

```
sum = 0
for i in range(1, num+1):
    sum = sum + i
    print(sum)
```

Sum = 0
0+1
1+2
1+3
3

o/p: enter a no: 6

1
3
6
10
15

for i in range(1, num+1):

5) Write a Program to count number of odd numbers from 1 to 30

count = 0 - if u want total count u should initialise any variable to zero & start doing this things

```
for i in range(1, 31, 1):
    if (i % 2 == 1):
        count = count + 1
    else:
        count = count
print(count)
```

o/p: 15

count = 0

- Start a counter at 0
- This variable will store how many odd no we find.

for i in range(1, 31, 1):

range(start, stop, step) → here

So it produces no
1, 2, 3, 4, ... 29, 30

this loop will run 30 times with i having each value

if $i \% 2 == 1$:

$i \% 2$ is modulo $\rightarrow$ gives the remainder after division

if $i \% 2 == 1$, then i is odd

if $i \% 2 == 0$, then i is even

Count = count + 1 (when i is odd)

• Increase count by 1 each time an odd number is found.

Ex:- $i = 1 \rightarrow$ odd $\rightarrow$ count becomes 1

$i = 2 \rightarrow$ even $\rightarrow$ count stays 1

$i = 3 \rightarrow$ odd $\rightarrow$ count becomes 2

continues like this until $i = 30$

else: count = count

if i is even, don't change anything

Actually this line is not needed, because leaving count as is has the same effect.

you could simply write:

if $i \% 2 == 1$:

count = count + 1

In short:

Loop goes through numbers 1 to 30

Each time it finds an odd number, count goes up by 1

Final result: 15

count = 0

for i in range(1, 31, 1):

if $i \% 2 == 1$:

count = count + 1

Print(count)

} checking & counting

o/p: 15

count = 0

for i in range(1, 31, 2):

count = count + 1

Print(count)

} directly counting
for every odd no. just incrementing count

o/p: 15

1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29

For loop repeats upto 15 times

It's just a count that is many times the loop has repeated for odd no.

6) Write a program to calculate sum of values of given range

```
# take start and end values from the user  
start = int(input("Enter start of range:"))  
end = int(input("Enter end of range:"))
```

```
# initialize sum
```

```
total = 0
```

```
# Loop from start to end (inclusive)
```

```
for i in range(start, end + 1):
```

```
    total += i # same as total = total + i
```

```
Print(f"Sum of values from {start} to {end} is {total}")
```

o/p: Enter start of range: 1

Enter end of range: 10

Loop will add:

$1 + 2 + 3 + \dots + 10 = 55$

Sum of values from 1 to 10 is 55

7) Write a Program to count total number of values, num of even, num of odd within a given range

1 to 10

10 - no of values

5 - no of even no

5 - no of odd no

→ `sn = int(input())`

`en = int(input())`

`nc = 0`

`ec = 0`

`oc = 0`

```
for i in range(sn, en + 1):
```

```
    nc = nc + 1 # i = 1, 2, 3, ..., 10
```


if $(i \% 2 == 1)$:

$oc = oc + 1$ # $i = 1, 3, 5, 7, 9$

else:

$ec = ec + 1$ # $i = 2, 4, 6, 8, 10$

Print (" num of values {nc} / num of even {ec} /
num of odd {oc} ")

o/p:

10

num of values 10

num of even 5

num of odd 5

• start : $nc = 0, ec = 0, oc = 0$

$i = 1 \rightarrow nc = 1, oc = 1, ec = 0$

$i = 2 \rightarrow nc = 2, oc = 1, ec = 1$

$i = 3 \rightarrow nc = 3, oc = 2, ec = 1$

$i = 4 \rightarrow nc = 4, oc = 2, ec = 2$

$i = 10 \rightarrow nc = 10, oc = 5, ec = 5$

final count : $nc = 10, ec = 5, oc = 5$

Prime No.

Any No. which is
divisible by itself
and 1 is called

Prime No.